

04

FOR IBPS SBI PO/CLERK PRELIMS 2026

QUANT CHECKLIST

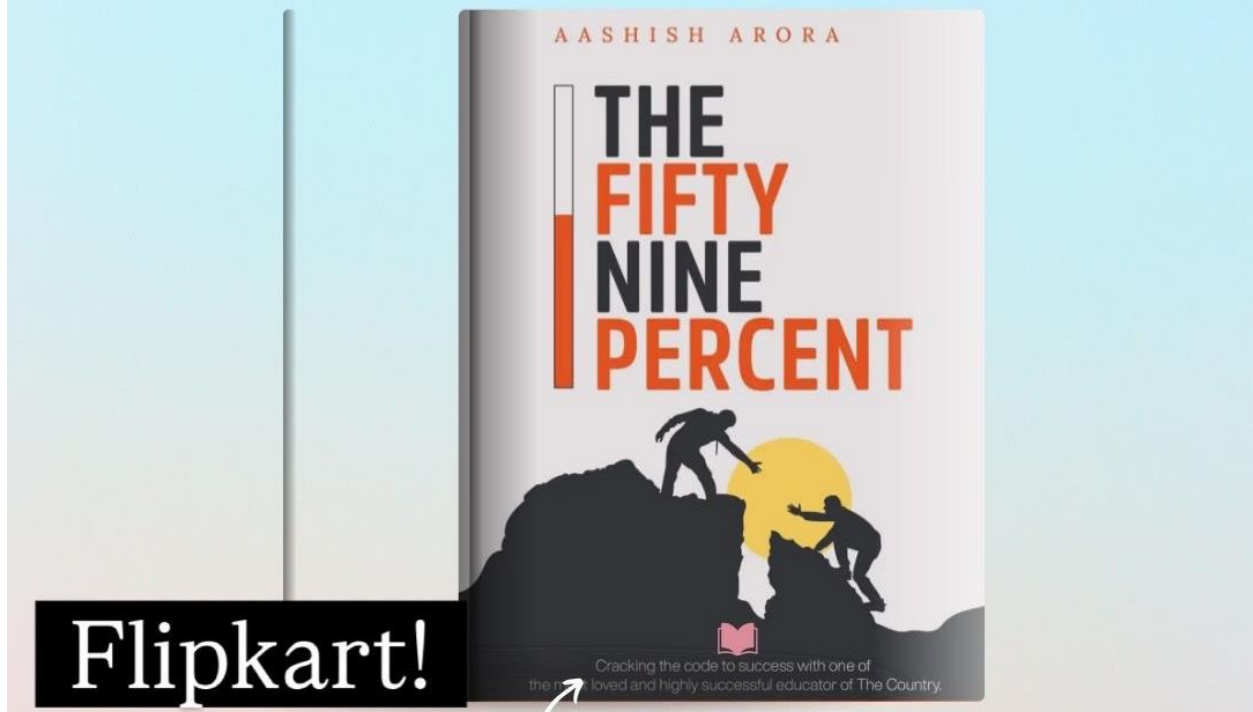
# *the* PRACTICE PAPER



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# COMPLETE

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**17 th FEBRUARY**

**BY AASHISH ARORA**

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Dear Students,

This daily practice sheet is designed to build both speed and accuracy, one day at a time.

It contains a mix of easy, moderate, and challenging questions to prepare you for every possible scenario in the exam. Treat it like a warm-up before the real game.

Solve it daily without fail. Don't wait for motivation—show up with discipline. Because it's not talent but consistent hard work that takes you places.

Stay focused. Stay consistent. Work hard.

-Aashish Arora

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# 1. SIMPLIFICATION AND APPROXIMATION

**Direction:** What value should come in place of the question mark (?) in the following question?

1.  $\sqrt{x} \times (35\% \text{ of } 420 + 975 \div 13) - (45)^2 = 417$

- a.100
- b.121
- c.144
- d.169
- e.196

2.  $600 \div \sqrt{576} + (16)^2 + 45\% \text{ of } 160 = 5x$

- a.73.6
- b.70.6
- c.67.6
- d.64.6
- e.61.6

3.  $(32 \times 11) + (42 \times 2.5) + (28 \times 3.75) = (x)^3 + 219$

- a.9
- b.10

- c.6
- d.7
- e.8

4.  $(2415 + 1455) \div 18 + (1425 + 2180) \div 35 = 6x - (6)^3$

- a.73
- b.77
- c.81
- d.85
- e.89

5.  $574 \div 14 \times 11 + (111 \times 15) = (112.5 \times 16) + 4x$

- a.79
- b.81
- c.83
- d.85
- e.87

6.  $(\sqrt[3]{2744} \times \sqrt[3]{5832} \times \sqrt{729}) \div 14 - (2.25 \times 48) = 6x$

- a.63
- b.65
- c.67

d.69

e.71

$$7. (125 \times 14) - 65\% \text{ of } 960 - 85\% \text{ of } 480 = 5x + (11 \times 48)$$

a.35

b.36

c.37

d.38

e.39

$$8. (112.5\% \text{ of } 960) \div 24 + (17)^3 = (65)^2 + (27)^2 + (x)^2$$

a.2

b.3

c.4

d.5

e.6

$$9. (x\% \text{ of } 560 + 8 \times 4096^{0.25}) \div 8 + (9.25 + 3.65 + 4.6) = 36$$

a.18

b.21

c.24

d.12

e.15

$$10. 40\% \text{ of } (20^2 + 25^2 + 35^2) = 75\% \text{ of } 980 + 15x$$

a.10

b.11

c.12

d.13

e.14

$$11. \sqrt{1156} + \sqrt{676} + \sqrt{1521} + (9)^3 = 42\% \text{ of } 1450 + 3x$$

a.71

b.73

c.75

d.77

e.79

$$12. (32)^2 - (24)^2 + 32\% \text{ of } 450 = 6x + (2.5 \times 112)$$

a.40

b.44

c.48

d.52

e.56

$$13. (3269 + 6598 + 4693) \div (28 \times 8) = 6x + \sqrt[3]{12167}$$

a.7

b.9

c.11

d.13

e.15

$$14.(11 \times 33) + (111 \times 39) - (45 \times 12) = (x)^2 + 3896$$

a.32

b.28

c.24

d.20

e.16

$$15.11.11\% \text{ of } 12.5\% \text{ of } 37.5\% \text{ of } 44.44\% \text{ of } 10368 = x$$

a.12

b.18

c.24

d.30

e.36

$$16.1024 \div 1.6 \div 6.4 \div 2.5 + (1296)^{\frac{1}{4}} = x$$

a.42

b.46

c.50

d.54

e.58

$$17.(x\% \text{ of } 1450) + (1344 \div 24) = 25\% \text{ of } 1616$$

a.12

b.16

c.20

d.24

e.28

$$18.(33)^2 + (36)^2 - (31)^2 = 9x + (14 \times 76)$$

a.40

b.48

c.56

d.64

e.72

$$19.35\% \text{ of } (960 + 1480 - 340) - (25)^2 = 11x$$

a.25

b.30

c.10

d.15

e.20

$$20.(35\% \text{ of } 28^2) + (40\% \text{ of } 24^2) = 4x + (55\% \text{ of } 16^2)$$

a.67

b.73

c.79

d.85

e.91

1. (b)



2. (b)

3. (d)

4. (e)

5. (a)

6. (a)

7. (d)

8. (a)

9. (e)

10.(b)

11.(b)

12.(d)

13.(a)

14.(e)

15.(c)

16.(b)

17.(d)

18.(a)

19.(c)

20.(e)

$$1) \sqrt{x} * \left( \frac{35}{100} * 420 + \frac{975}{13} \right) - 2025 = 417$$

$$\sqrt{x} * (147 + 75) - 2025 = 417$$

$$\sqrt{x} * 222 = 417 + 2025$$

$$\sqrt{x} = \frac{2442}{222}$$

$$\sqrt{x} = 11 = 121$$

$$2) \frac{600}{24} + 256 + \frac{45}{100} * 160 = 5x$$

$$25 + 256 + 72 = 5x$$

$$353 = 5x$$

$$x = \frac{353}{5} = 70.6$$

$$3) 352 + 105 + 105 = x^3 + 219$$

$$562 = x^3 + 219$$

$$562 - 219 = x^3$$

$$x^3 = 343 = 7$$

$$4) \frac{3870}{18} + \frac{3605}{35} = 6x - 216$$

$$215 + 103 = 6x - 216$$

$$318 = 6x - 216$$

$$318 + 216 = 6x$$

$$\frac{534}{6} = x = 89$$

$$5) \frac{574}{14} * 11 + 1665 = 1800 + 4x$$

$$41 * 11 + 1665 = 1800 + 4x$$

$$451 + 1665 = 1800 + 4x$$

$$2116 - 1800 = 4x$$

$$\frac{316}{4} = x = 79$$

$$6) \frac{14 \cdot 18 \cdot 27}{14} - 108 = 6x$$

$$\frac{6804}{14} - 108 = 6x$$

$$486 - 108 = 6x$$

$$\frac{378}{6} = x = 63$$

$$7) 1750 - \frac{65}{100} * 960 - \frac{85}{100} * 480 = 5x + 528$$

$$1750 - 624 - 408 = 5x + 528$$

$$718 - 528 = 5x$$

$$\frac{190}{5} = x = 38$$

$$8) \left( \frac{1125}{100 \cdot 10} * 960 \right) \div 24 + 4913 = 4225 + 729 + x^2$$

$$\frac{1080}{24} + 4913 = 4954 + x^2$$

$$45 + 4913 = 4954 + x^2$$

$$4958 - 4954 = x^2$$

$$4 = x^2 = 2$$

$$9) \left( \frac{x}{100} * 560 + 8 * 8 \right) \div 8 + 17.5 = 36$$

$$\frac{\frac{56x}{10} + 64}{8} + 17.5 = 36$$

$$\frac{\frac{56x}{10} + 64}{8} = 36 - 17.5$$

$$\frac{56x}{10} + 64 = 18.5 * 8$$

$$\frac{56x}{10} = 148 - 64$$

$$\frac{56x}{10} = 84$$

$$x = \frac{84 \cdot 10}{56} = 15$$

$$10) \frac{40}{100} * (400 + 625 + 1225) = \frac{75}{100} * 980 + 15x$$

$$\frac{40}{100} * 2250 = 735 + 15x$$

$$900 - 735 = 15x$$

$$\frac{165}{15} = x = 11$$

$$11) 34 + 26 + 39 + 729 = \frac{42}{100} * 450 + 3x$$

$$828 = 609 + 3x$$

$$3x = 828 - 609$$

$$x = \frac{219}{3} = 73$$

$$12) 1024 - 576 + \frac{32}{100} * 450 = 6x + 280$$

$$448 + 144 = 6x + 280$$

$$592 - 280 = 6x$$

$$\frac{312}{6} = x = 52$$

$$13) \frac{14560}{224} = 6x + 23$$

$$65 - 23 = 6x$$

$$\frac{42}{6} = x = 7$$

$$14) 363 + 4329 - 540 = x^2 + 3896$$

$$4692 - 540 = x^2 + 3896$$

$$4152 - 3896 = x^2$$

$$x^2 = 256 = 16$$

$$15) \quad \frac{1}{9} * \frac{1}{8} * \frac{3}{8} * \frac{4}{9} * \\ 10368 = x$$

$$\frac{10368 * 12}{5184} = x$$

$$x = 24$$

$$16) \quad \frac{1024}{1.6} * \frac{1}{6.4} * \frac{1}{2.5} + \\ (6^4)^{\frac{1}{4}} = x$$

$$\frac{640}{6.4} * \frac{1}{2.5} + 6 = x$$

$$40 + 6 = x = 46$$

$$17) \quad \frac{x}{100} * 1450 + \\ \frac{1344}{24} = \frac{25}{100} * 1616$$

$$\frac{145x}{10} + 56 = 404$$

$$\frac{145x}{10} = 404 - 56$$

$$x = \frac{348 * 10}{145}$$

$$x = 24$$

$$18) \quad 1089 + 1296 - \\ 961 = 9x + 1064$$

$$2385 - 961 = 9x + 1064$$

$$1424 - 1064 = 9x$$

$$\frac{360}{9} = x = 40$$

$$19) \quad \frac{35}{100} * 2100 - \\ 625 = 11x$$

$$735 - 625 = 11x$$

$$\frac{110}{11} = x = 10$$

$$20) \quad \frac{35}{100} * 784 + \frac{40}{100} * \\ 576 = 4x + \frac{55}{100} * 256$$

$$274.4 + 230.4 = 4x + 140.8$$

$$504.8 - 140.8 = 4x$$

$$\frac{364}{4} = x = 91$$



## **FOUND ERROR?**

Report the error in the checklist to  
**[teamchecklist22@gmail.com](mailto:teamchecklist22@gmail.com)**

Adapted from  
Kern, 2014a

PRACTICE PAIR

## 2. ARITHMETIC QUESTIONS

(1) In a class there are 16 boys and 14 girls. The average weight of each boy and each girl is 48 kg and 44 kg. If the sum of weight of all the students and a teacher is 1449 kg, then find the weight of the teacher.

एक क्लास में 16 लड़के और 14 लड़कियाँ हैं। हर लड़के और हर लड़की का औसत वज़न 48 kg और 44 kg है। अगर सभी स्टूडेंट्स और एक टीचर के वज़न का कुल योग 1449 kg है, तो टीचर का वज़न पता करें।

- (a) 54 kg
- (b) 65 kg
- (c) 52 kg
- (d) 68 kg
- (e) None of these

(2) A train crosses a bridge in 23 seconds while it crosses a man standing on a platform in 17 seconds. If the speed of the train is 126 km/hour, find the time taken by the train to cross a tunnel whose length is 175 metres more than the length of the bridge.

एक ट्रेन एक पुल को 23 सेकंड में पार करती है, जबकि वह प्लेटफॉर्म पर खड़े एक आदमी को 17 सेकंड में पार करती है। अगर ट्रेन की स्पीड 126 km/घंटा है, तो उस सुरंग को पार करने में ट्रेन को कितना समय लगेगा जिसकी लंबाई पुल की लंबाई से 175 मीटर ज़्यादा है?

- (a) 36 Seconds
- (b) 42 Seconds

- (c) 28 Seconds
- (d) 32 Seconds
- (e) None of these

(3) Anshul started a business by investing Rs 7200. After some months, Deepu joined him with Rs 6400. At the end of the year, out of the total profit of Rs 17500, Anshul received Rs 13500, after how many months Deepu joined the business.

अंशुल ने 7200 रुपये लगाकर एक बिज़नेस शुरू किया। कुछ महीनों बाद, दीपू 6400 रुपये लगाकर उसके साथ जुड़ गया। साल के आखिर में, 17500 रुपये के कुल मुनाफ़े में से अंशुल को 13500 रुपये मिले, तो दीपू कितने महीनों बाद बिज़नेस में शामिल हुआ?

- (a) 5 Months
- (b) 6 Months
- (c) 9 Months
- (d) 8 Months
- (e) None of these

(4) A jar contains 240 litres of a mixture of wine and water in the ratio 8:7 respectively. If X litres of wine is added to the mixture, then the ratio of wine to water in the resultant mixture becomes 22:7. Find the value of X.

एक जार में वाइन और पानी का 240 लीटर मिश्रण है, जिसका अनुपात क्रमशः 8:7 है। अगर मिश्रण में x लीटर वाइन मिलाई जाती है, तो बनने वाले मिश्रण में वाइन और पानी का अनुपात 22:7 हो जाता है। x का मान ज्ञात कीजिए।

- (a) 258 litres
- (b) 224 liters

- (c) 265 litres
- (d) 286 liters
- (e) None of these

(5) Rs 5600 when invested at SI of 15% per annum for some years gives an interest of Rs 4200. If Rs 8000 is invested at CI compounded annually of 10% per annum for 3 years less than the time for which Rs 5600 was invested. Find the compound interest earned on Rs 8000.

जब 5600 रुपये को कुछ सालों के लिए 15% प्रति वर्ष की साधारण ब्याज दर पर इन्वेस्ट किया जाता है, तो 4200 रुपये का ब्याज मिलता है। अगर 8000 रुपये को 10% प्रति वर्ष की चक्रवृद्धि ब्याज दर पर, उस समय से 3 साल कम समय के लिए इन्वेस्ट किया जाता है, जितने समय के लिए 5600 रुपये इन्वेस्ट किए गए थे, तो 8000 रुपये पर मिलने वाला चक्रवृद्धि ब्याज ज्ञात कीजिए।

- (a) 1680
- (b) 1240
- (c) 1620
- (d) 1280
- (e) None of these

(6) The average age of A, B and C four years ago was 32 years, and the average of the present ages of A and B is 28 years. The ratio of the ages of C and D is 4:3. Find the age of D after 6 years.

चार साल पहले A, B और C की उम्र का औसत 32 साल था, और A और B की वर्तमान उम्र का औसत 28 साल है। C और D की उम्र का अनुपात 4:3 है। 6 साल बाद D की उम्र ज्ञात कीजिए।

- (a) 82 years

- (b) 32 years
- (c) 45 years
- (d) 56 years
- (e) None of these

(7) An article was marked 50% above its cost price and sold after two successive discounts of 10% and 20% respectively. If the profit earned by the shopkeeper is Rs 900, find the cost price of the article.

एक आर्टिकल पर उसकी लागत कीमत से 50% ज़्यादा कीमत मार्क की गई और उसे 10% और 20% के दो लगातार डिस्काउंट के बाद बेचा गया। अगर दुकानदार को Rs 900 का प्रॉफ़िट हुआ, तो आर्टिकल की लागत कीमत पता करें।

- (a) 11250
- (b) 15420
- (c) 16520
- (d) 14250
- (e) None of these

(8) The circumference of a circle is 220 cm and the radius of the circle is 4 cm more than the side of a square. Find the area of the square.

एक वृत्त की परिधि 220 cm है और वृत्त की त्रिज्या एक वर्ग की भुजा से 4 cm ज़्यादा है। वर्ग का क्षेत्रफल ज्ञात कीजिए।

- (a)  $529 \text{ cm}^2$
- (b)  $784 \text{ cm}^2$
- (c)  $441 \text{ cm}^2$



(d)  $961 \text{ cm}^2$

(e) None of these

(9) Abhi and Bini together can complete a piece of work in 24 days. Chandan and Abhi can complete the same work in 30 days. If Chandan alone can complete the same work in 60 days, then Bini and Chandan can complete 37.5% of the work in how many days?

अभि और बिनी मिलकर एक काम को 24 दिनों में पूरा कर सकते हैं। चंदन और अभि उसी काम को 30 दिनों में पूरा कर सकते हैं। अगर चंदन अकेला उसी काम को 60 दिनों में पूरा कर सकता है, तो बिनी और चंदन मिलकर काम का 37.5% कितने दिनों में पूरा करेंगे?

(a) 16 days

(b) 9 days

(c) 12 days

(d) 4 days

(e) None of these

(10) The speed of the boat in still water is  $15 \text{ km/hour}$  while its speed downstream is  $23 \text{ km/hour}$ . Find the time taken by the boat to cover  $126 \text{ km}$  distance upstream.

शांत पानी में नाव की स्पीड  $15 \text{ km/hour}$  है, जबकि धारा के अनुकूल इसकी स्पीड  $23 \text{ km/hour}$  है। नाव को धारा के प्रतिकूल  $126 \text{ km}$  की दूरी तय करने में कितना समय लगेगा?

(a) 17 hours

(b) 12 hours

(c) 18 hours

(d) 15 hours

(e) None of these

(11) Anu spends  $\frac{1}{5}$  th of his monthly salary on house rent,  $\frac{1}{8}$  th of the remaining on food and half of the remaining on entertainment. If the difference between the amount he spends on rent and that on entertainment is Rs.2700, then find the monthly salary of Anu?

अनु अपनी महीने की सैलरी का  $\frac{1}{5}$  हिस्सा घर के किराए पर, बाकी का  $\frac{1}{8}$  हिस्सा खाने पर और बचे हुए का आधा हिस्सा एंटरटेनमेंट पर खर्च करता है। अगर किराए और एंटरटेनमेंट पर खर्च की गई रकम के बीच का अंतर 2700 रुपये है, तो अनु की महीने की सैलरी कितनी है?

(a) 12000

(b) 14000

(c) 15000

(d) 18000

(e) None of these

(12) Train A can cross a pole and 60 meters long bridge in 20 seconds and 32 seconds respectively. The length of train B is 320 meters and it is running at the speed of 54 km/hr. If both the trains A and B run in opposite directions and cross each other in "x" seconds, then find the value of "x"?

ट्रेन A एक पोल और 60 मीटर लंबे पुल को क्रमशः 20 सेकंड और 32 सेकंड में पार कर सकती है। ट्रेन B की लंबाई 320 मीटर है और यह 54 km/hr की स्पीड से चल रही है। अगर दोनों ट्रेनें A और B विपरीत दिशा में चल रही हैं और एक-दूसरे को "x" सेकंड में पार करती हैं, तो "x" का मान ज्ञात करें?

(a) 24 Seconds

(b) 22 Seconds

- (c) 21 Seconds
- (d) 28 Seconds
- (e) None of these

(13) The ratio of the age of Arpit and Bhanu is 4:7 respectively. After 12 years, the ratio of their ages will be 8:11 respectively. If the age of Krishna is 25% more than the age of Arpit, then find the age of Krishna?

अर्पित और भानु की उम्र का अनुपात क्रमशः 4:7 है। 12 साल बाद, उनकी उम्र का अनुपात क्रमशः 8:11 होगा। अगर कृष्णा की उम्र अर्पित की उम्र से 25% ज़्यादा है, तो कृष्णा की उम्र पता करें?

- (a) 12 years
- (b) 15 years
- (c) 16 years
- (d) 18 years
- (e) None of these

(14) The speed of a boat in still water is 36 km/hr. If the speed of the boat upstream is 17 km/hr, then how many hours will it take to travel 935 km downstream?

शांत पानी में नाव की स्पीड 36 km/hr है। अगर धारा के विपरीत नाव की स्पीड 17 km/hr है, तो धारा की दिशा में 935 km की दूरी तय करने में कितना समय लगेगा?

- (a) 19 hours
- (b) 11 hours
- (c) 14 hours
- (d) 17 hours

(e) None of these

(15) A shopkeeper has 1170 kg of sugar, part of which he sells at 55% profit and the rest at 68% profit. He gained 62% on the whole. Find the quantity (in kg) sold at 55% profit?

एक दुकानदार के पास 1170 kg चीनी है, जिसका कुछ हिस्सा वह 55% मुनाफे पर और बाकी 68% मुनाफे पर बेचता है। उसे कुल मिलाकर 62% का मुनाफा हुआ। 55% मुनाफे पर बेची गई चीनी की मात्रा (kg में) ज्ञात कीजिए?

(a) 540 kg

(b) 620 kg

(c) 580 kg

(d) 640 kg

(e) None of these

(16) In Vani purse, there are 828 coins of 50p, 20p and 1Rs. The ratio of their value is 8:12:16. Calculate the number of 20 paise coins?

वाणी के पर्स में 50 पैसे, 20 पैसे और 1 रुपये के 828 सिक्के हैं। उनकी कीमत का अनुपात 8:12:16 है। 20 पैसे के सिक्कों की संख्या ज्ञात कीजिए?

(a) 460

(b) 320

(c) 540

(d) 620

(e) None of these

(17) A class has 6 boys and 4 girls. In how many ways can a team of 4 students be selected such that exactly 2 girls are included?

एक क्लास में 6 लड़के और 4 लड़कियाँ हैं। 4 स्टूडेंट्स की एक टीम कितने तरीकों से चुनी जा सकती है जिसमें ठीक 2 लड़कियाँ हों?

- (a) 50
- (b) 90
- (c) 40
- (d) 80
- (e) None of these

(18) If the average weight of "x" girls in an office is 48 kg and total weight of all employees (boys and girls) together are 1584 kg. If the average weight of 16 boys is 54 kg, then find the value of 'X' ?

अगर एक ऑफिस में "x" लड़कियों का औसत वज़न 48 kg है और सभी कर्मचारियों (लड़के और लड़कियों) का कुल वज़न 1584 kg है। अगर 16 लड़कों का औसत वज़न 54 kg है, तो 'x' का मान ज्ञात कीजिए?

- (a) 17
- (b) 18
- (c) 12
- (d) 15
- (e) None of these

(19) Bitu cycles with an average speed of 27 km/hr. He takes 140 minutes to reach the zoo from his home. Then find how far is the zoo from his home?

बिटू 27 km/hr की औसत स्पीड से साइकिल चलाता है। उसे अपने घर से चिड़ियाघर पहुँचने में 140 मिनट लगते हैं। तो पता लगाएँ कि चिड़ियाघर उसके घर से कितनी दूर है?

- (a) 63 km
- (b) 57 km
- (c) 78 km
- (d) 97 km
- (e) None of these

(20) The sum between 75% of a number and 20% of the same number is 855. What is the sum of all the digits of the number?

किसी संख्या के 75% और उसी संख्या के 20% का योग 855 है। उस संख्या के सभी अंकों का योग क्या है?

- (a) 7
- (b) 8
- (c) 9
- (d) 5
- (e) None of these

Answers:

- (1) B
- (2) C
- (3) D
- (4) B

- (5) A
- (6) C
- (7) A
- (8) D
- (9) B
- (10) C

(11) D

(12) C

(13) B

(14) D

(15) A

(16) C

(17) B

(18) D

(19) A

(20) C

Solutions:

1) Number of boys = 16

Average weight of each boy = 48 kg

Total weight of boys

$$= 16 \times 48$$

$$= 768 \text{ kg}$$

Number of girls = 14

Average weight of each girl = 44 kg

Total weight of girls

$$= 14 \times 44$$

$$= 616 \text{ kg}$$

Total weight of all students

$$= 768 + 616$$

$$= 1384 \text{ kg}$$

Given, total weight of all students and the teacher = 1449 kg

Weight of the teacher

$$= 1449 - 1384$$

$$= 65 \text{ kg Ans}$$

2) Speed of the train = 126 km/hour

Convert into m/s

$$= 126 \times \frac{5}{18}$$

$$= 35 \text{ m/s}$$

Time taken to cross a man = 17sec

So, length of the train

$$= \text{speed} \times \text{time}$$

$$= 35 \times 17$$

$$= 595 \text{ m}$$

Time taken to cross the bridge = 23sec

So, length of train + length of bridge

$$= 35 \times 23$$

$$= 805 \text{ m}$$

Length of the bridge

$$= 805 - 595$$

$$= 210 \text{ m}$$

Length of the tunnel

$$= \text{length of bridge} + 175$$

$$= 210 + 175$$

$$= 385 \text{ m}$$

Now, length of train + length of tunnel

$$= 595 + 385$$

$$= 980 \text{ m}$$

Time taken to cross the tunnel



= total length / speed

=  $980 / 35$

= 28 seconds Ans

3) Investment of Anshul = 7200

Investment of Deepu = 6400

Total profit = 17500

Anshul's share = 13500

Deepu's share

=  $17500 - 13500$

= 4000

Profit ratio

Anshul : Deepu

=  $13500 : 4000$

=  $27 : 8$

Anshul invested for 12 months

Let Deepu invest for  $t$  months

So,

$7200 \times 12 : 6400 \times t = 27 : 8$

$86400 : 6400t = 27 : 8$

$86400 \times 8 = 27 \times 6400t$

$691200 = 172800t$

$t = 4$  months

So, Deepu invested his money for 4 months

Time after which Deepu joined

$$= 12 - 4$$

= 8 months Ans

4) The initial quantities of wine and water

$$\text{Wine} = (8/15) \times 240 = 128 \text{ litres}$$

$$\text{Water} = (7/15) \times 240 = 112 \text{ litres}$$

Let X litres of wine be added

Then the new quantity of wine =  $128 + X$  litres

Water remains 112 litres

New ratio of wine to water = 22:7

$$\text{So, } (128 + X)/112 = 22/7$$

$$7 \times (128 + X) = 22 \times 112$$

$$896 + 7X = 2464$$

$$7X = 2464 - 896$$

$$7X = 1568$$

$$X = 1568 / 7$$

$$X = 224 \text{ litres Ans}$$

$$5) 4200 = (5600 \times 15 \times T)/100$$

$$4200 = 840 \times T$$

$$T = 4200 / 840 = 5 \text{ years}$$

$$\text{Time} = 5 - 3 = 2 \text{ years}$$

For 2 years at 10% per annum:

$$CI = P \times (R + (R \times R)/100)/100$$

Here, R = 10%

$$CI = 8000 \times \{10 + 10(10 \times 10)/100\}/100$$

$$CI = 8000 \times 21/100 = 1680 \text{ Ans}$$

6) Total age of A, B, C four years ago =  $32 \times 3 = 96$  years

Let present ages of A, B, C be A, B, C respectively.

Four years ago:

$$(A - 4) + (B - 4) + (C - 4) = 96$$

$$A + B + C - 12 = 96$$

$$A + B + C = 108$$

The present average of A and B is 28 years:

$$(A + B)/2 = 28$$

$$A + B = 56$$

$$C = 108 - 56 = 52$$

The ratio of C and D is 4:3

$$D = (3/4) \times C = (3/4) \times 52 = 39$$

Age of D after 6 years

$$= 39 + 6 = 45 \text{ years Ans}$$

7) MP is 50% above CP

$$CP : MP = 100 : 150 = 2 : 3$$

First discount 10%

$$SP1 : MP = 90 : 100 = 9 : 10$$

Second discount 20%

$$SP : SP1 = 80 : 100 = 4 : 5$$

Combine ratios:

$$SP : CP = SP : SP1 \times SP1 : MP \times MP : CP$$

$$SP : CP = (4/5) \times (9/10) \times (3/2)$$

$$SP : CP = (4 \times 9 \times 3) / (5 \times 10 \times 2) = 108 / 100 = 1.08$$

$$\text{Profit} = SP - CP = (1.08 - 1) \times CP = 0.08 \times CP$$

$$\text{Given profit} = 900$$

$$0.08 \times CP = 900$$

$$CP = 900 / 0.08 = 11250 \text{ Ans}$$

$$8) 220 = 2 \times (22/7) \times r$$

$$220 = (44/7) \times r$$

$$r = (220 \times 7) / 44$$

$$r = 1540 / 44 = 35 \text{ cm}$$

$$\text{Let the side of the square} = a \text{ cm}$$

$$r = a + 4$$

$$35 = a + 4$$

$$a = 35 - 4 = 31 \text{ cm}$$

$$\text{Area of square} = a^2$$

$$= 31^2 = 961 \text{ cm}^2 \text{ Ans}$$

$$9) \text{ LCM of } 24, 30, 60 = 120 \text{ units of work}$$

$$\text{Abhi + Bini} = 120/24 = 5 \text{ units per day}$$

$$\text{Chandan + Abhi} = 120/30 = 4 \text{ units per day}$$

$$\text{Chandan alone} = 120/60 = 2 \text{ units per day}$$

$$\text{Abhi} = (\text{Chandan + Abhi}) - \text{Chandan} = 4 - 2 = 2 \text{ units per day}$$

$$\text{Bini} = (\text{Abhi + Bini}) - \text{Abhi} = 5 - 2 = 3 \text{ units per day}$$

Bini + Chandan =  $3 + 2 = 5$  units per day

37.5% of work =  $\frac{3}{8} \times 120 = 45$  units

Time =  $45 / 5 = 9$  days Ans

10) Speed of boat in still water = 15 km/h

Downstream speed = 23 km/h

Speed of current =  $23 - 15 = 8$  km/h

Upstream speed =  $15 - 8 = 7$  km/h

Distance upstream = 126 km

Time = Distance / Speed =  $126 / 7 = 18$  hours Ans

11) Let the monthly salary of Anu = ₹x

Amount spent on house rent

=  $(\frac{1}{5})x$

Remaining salary after rent

=  $x - (\frac{1}{5})x = (\frac{4}{5})x$

Amount spent on food

=  $(\frac{1}{8}) \times (\frac{4}{5})x = (\frac{1}{10})x$

Remaining salary after food

=  $(\frac{4}{5})x - (\frac{1}{10})x = (\frac{7}{10})x$

Amount spent on entertainment

=  $(\frac{1}{2}) \times (\frac{7}{10})x = (\frac{7}{20})x$

Given difference between rent and entertainment = 2700

$(\frac{7}{20})x - (\frac{1}{5})x = 2700$

$(\frac{7}{20})x - (\frac{4}{20})x = 2700$

$$(3/20)x = 2700$$

$$x = (2700 \times 20) / 3$$

$$x = 18000 \text{ Ans}$$

12) Time taken by train A to cross a pole = 20 sec

Length of train A / Speed of train A = 20

Time taken by train A to cross a 60 m bridge = 32 sec

(Length of train A + 60) / Speed of train A = 32

Let length of train A = L and speed = v

$$L / v = 20$$

$$L = 20v$$

$$(L + 60) / v = 32$$

$$= (20v + 60) / v = 32$$

$$= 20 + 60/v = 32$$

$$= 60/v = 12$$

$$v = 5 \text{ m/sec}$$

So, length of train A

$$L = 20 \times 5 = 100 \text{ m}$$

Speed of train B = 54 km/hr

$$= 54 \times (5/18) = 15 \text{ m/sec}$$

Length of train B = 320 m

Relative speed = 5 + 15 = 20 m/s

Total distance = 100 + 320 = 420 m

Time = Distance / Relative speed

$$x = 420 / 20$$

$x = 21$  seconds Ans

13) Let the present ages of Arpit and Bhanu be  $4x$  and  $7x$  respectively

After 12 years,

Age of Arpit =  $4x + 12$

Age of Bhanu =  $7x + 12$

Given ratio after 12 years

$$(4x + 12) / (7x + 12) = 8 / 11$$

$$11(4x + 12) = 8(7x + 12)$$

$$44x + 132 = 56x + 96$$

$$12x = 36$$

$$x = 3$$

Present age of Arpit

$$= 4x = 4 \times 3 = 12 \text{ years}$$

Age of Krishna is 25% more than age of Arpit

$$= 125\% \text{ of } 12$$

$$= (125/100) \times 12$$

$$= 15 \text{ years years Ans}$$

14) Speed of boat in still water = 36 km/hr

Speed of boat upstream = 17 km/hr

$$\text{Speed of stream} = 36 - 17 = 19 \text{ km/hr}$$

Speed of boat downstream

$$= 36 + 19 = 55 \text{ km/hr}$$

Distance downstream = 935 km

Time taken

$$= 935 / 55$$

= 17 hours Ans

15) By allegation method,

Ratio of quantities

$$= (68 - 62) : (62 - 55)$$

$$= 6 : 7$$

Total quantity = 1170 kg

$$\text{Sum of ratio} = 6 + 7 = 13$$

$$13 \text{ units} = 1170$$

$$1 \text{ unit} = 1170 / 13 = 90$$

Quantity sold at 55% profit

$$= 6 \text{ units}$$

$$= 6 \times 90 = 540 \text{ kg Ans}$$

16) 50 paise = 2 coin

20 paise = 5 coin

1 Rs = 1 coin

50 paise coins

$$= 8 \times 2 = 16 \text{ units}$$

20 paise coins

$$= 12 \times 5 = 60 \text{ units}$$

1 rupee coins

$$= 16 \times 1 = 16 \text{ units}$$



Total units

$$= 16 + 60 + 16 = 92 \text{ units}$$

$$92 \text{ units} = 828 \text{ coins}$$

$$1 \text{ unit} = 828 / 92 = 9 \text{ coins}$$

Number of 20 paise coins

$$= 60 \text{ units}$$

$$= 60 \times 9 = 540 \text{ coins Ans}$$

17) Number of girls = 4

Number of boys = 6

To form a team of 4 students with exactly 2 girls

Ways to select 2 girls out of 4

$$= {}^4C_2$$

Ways to select 2 boys out of 6

$$= {}^6C_2$$

Total number of ways

$$= {}^4C_2 \times {}^6C_2$$

$$= (4 \times 3) / (2 \times 1) \times (6 \times 5) / (2 \times 1)$$

$$= 6 \times 15$$

$$= 90 \text{ ways Ans}$$

18) Let the number of girls = x

Average weight of girls = 48 kg

Total weight of girls =  $48 \times x = 48x$

Number of boys = 16

Average weight of boys = 54 kg

Total weight of boys =  $16 \times 54 = 864$  kg

Total weight of all employees = 1584 kg

So,

$$48x + 864 = 1584$$

$$48x = 1584 - 864$$

$$48x = 720$$

$$x = 720 / 48$$

$$x = 15 \text{ Ans}$$

19) Average speed of Bitu = 27 km/hr

Time taken = 140 minutes

$$= 140 / 60 = 7/3 \text{ hours}$$

$$\text{Distance} = 27 \times 7/3 = 63 \text{ km Ans}$$

20) Let the number be x

$$75\% \text{ of the number} = (75/100)x = (3/4)x$$

$$20\% \text{ of the number} = (20/100)x = (1/5)x$$

Given sum

$$(3/4)x + (1/5)x = 855$$

$$(15x + 4x) / 20 = 855$$

$$19x / 20 = 855$$

$$19x = 855 \times 20$$

$$19x = 17100$$

$$x = 900$$

Sum of digits of the number

$$= 9 + 0 + 0$$

$$= 9 \text{ Ans}$$

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### 3. Quadratic Equations

In each of the following questions, there are two equations. You have to solve both equations and mark the correct answer.

(a)  $x > y$

(b)  $x < y$

(c)  $x = y$  or the relationship cannot be established

(d)  $x \geq y$

(e)  $x \leq y$

1.) I.  $x^2 - 28x + 171 = 0$

II.  $y^2 + 8y - 153 = 0$

2.) I.  $3x^2 + 2x - 56 = 0$

II.  $4y^2 - 34y + 72 = 0$

3.) I.  $x^2 - 20x + 75 = 0$

II.  $y^2 - 32y + 255 = 0$

4.) I.  $x^2 - 10x + 21 = 0$

II.  $y^2 - 7y + 12 = 0$

5.) I.  $5x^2 - 36x + 64 = 0$

II.  $2y^2 - 17y + 36 = 0$

6.) I.  $x^2 - 14x + 45 = 0$

II.  $y^2 - 33y + 216 = 0$

7.) I.  $x^2 - 22x + 105 = 0$

II.  $y^2 + 3y - 54 = 0$

8.) I.  $6x^2 + 11x - 72 = 0$

II.  $8y^2 - 52y + 84 = 0$

9.) I.  $x^2 - 13x + 61 = 25$

II.  $y^2 - 21y + 153 = 45$

10.) I.  $x^2 - 13x + 40 = 0$

II.  $y^2 - 18y + 81 = 0$

11.) I.  $16x^2 + 23x - 75 = 0$

II.  $8y^2 - 51y + 70 = 0$

12.) I.  $x^2 = 784$

II.  $y^2 - 57y + 812 = 0$

13.) I.  $x^2 + 9x - 36 = 0$

II.  $y^2 - 22y + 85 = 0$

14.) I.  $7x^2 + 26x - 96 = 0$

II.  $4y^2 - 30y + 56 = 0$

15.) I.  $x^3 = 8^3 - 387$

II.  $y = 4^2 - 12$

16.) I.  $x^2 = 225$

II.  $y^3 - 2051 = 2045$

17.) I.  $9x^2 - 46x + 56 = 0$

II.  $3y^2 - 24y + 48 = 0$

18.) I.  $x^2 - 33x + 272 = 0$

II.  $y^2 - 31y + 240 = 0$

19.) I.  $x^2 - 156 = 168$

II.  $y - \sqrt[3]{64} = \sqrt{289}$

20.) I.  $2x^2 - 20x + 48 = 0$

II.  $5y^2 - 45y + 100 = 0$

Answers:

1. D

2. E

3. E

4. C

5. E

6. E

7. A

8. B

9. E

10. B

11. B

12. E

13. B

14. B

15. A

16. B

17. B

18. D

19. B

20. C

Answers:

$$(1) x = 19,9$$

$$y = 9,-17$$

$$(2) x = 4,-14/3$$

$$y = 9/2,4$$

$$(3) x = 15,5$$

$$y = 17,15$$

$$(4) x = 7,3$$

$$y = 4,3$$

$$(5) x = 4,16/5$$

$$y = 9/2,4$$

$$(6) x = 9,5$$

$$y = 9,-24$$

$$(7) x = 15,7$$

$$y = 6,-9$$

$$(8) x = 8/3,-9/2$$

$$y = 7/2,3$$

$$(9) x = 9,4$$

$$y = 12,9$$

$$(10) x = 8,5$$

$$y = 9,9$$

$$(11) x = 25/16,-3$$

$$y = 35/8,2$$

$$(12) x = 28,-28$$

$$y = 29,28$$

$$(13) x = 3,-12$$

$$y = 17,5$$

$$(14) x = 16/7,-6$$

$$y = 4,7/2$$

$$(15) x = 5$$

$$y = 4$$

$$(16) x = 15,-15$$

$$y = 16$$

$$(17) x = 28/9,2$$

$$y = 4,4$$

$$(18) x = 17,16$$

$$y = 16,15$$

$$(19) x = 18,-18$$

$$y = 21$$

$$(20) x = 4,6$$

$$y = 5,4$$

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## 4. WRONG NUMBER SERIES

(1) 488, 430, 376, 336, 304, 278

- (a) 376
- (b) 278
- (c) 488
- (d) 304
- (e) None of these

(2) 10, 85, 172, 270, 382, 505

- (a) 172
- (b) 505
- (c) 85
- (d) 270
- (e) None of these

(3) 5, 200, 489, 889, 1418, 2094

- (a) 5
- (b) 1418
- (c) 889
- (d) 200
- (e) None of these

(4) 41, 95, 150, 209, 269, 331

- (a) 331
- (b) 150
- (c) 209
- (d) 269
- (e) None of these

(5) 72, 111, 158, 213, 276, 350

- (a) 111
- (b) 350
- (c) 213
- (d) 276
- (e) None of these

(6) 576, 545, 526, 520, 516, 521

- (a) 516
- (b) 576
- (c) 545
- (d) 521
- (e) None of these

(7) 20, 26, 54, 196, 958, 5724

- (a) 26



- (b) 958  
(c) 20  
(d) 54  
(e) None of these

(8) 4, 17, 60, 183, 560, 1693

- (a) 560  
(b) 60  
(c) 17  
(d) 183  
(e) None of these

(9) 222, 250, 218, 254, 214, 262

- (a) 262  
(b) 218  
(c) 214  
(d) 222  
(e) None of these

(10) 2, 4, 14, 129, 2068, 51705

- (a) 129  
(b) 2068  
(c) 4  
(d) 14  
(e) None of these

(11) 22, 31.5, 52.5, 92.5, 170.5, 324.5

- (a) 52.5  
(b) 170.5  
(c) 22  
(d) 92.5  
(e) None of these

(12) 2197, 170, 1352, 104, 832, 64

- (a) 832  
(b) 64  
(c) 1352  
(d) 170  
(e) None of these

(13) 84, 884, 1280, 1484, 1584, 1634

- (a) 1484  
(b) 884  
(c) 1634  
(d) 84  
(e) None of these

(14) 863, 694, 573, 495, 443, 418

- (a) 418  
(b) 495  
(c) 694  
(d) 863

(e) None of these

(15) 967, 945, 909, 857, 790, 697

(a) 790

(b) 967

(c) 697

(d) 909

(e) None of these

(16) 400, 382, 364, 346, 328, 312

(a) 328

(b) 312

(c) 400

(d) 364

(e) None of these

(17) 65, 69, 133, 202, 335, 537

(a) 537

(b) 335

(c) 65

(d) 69

(e) None of these

(18) 380016, 54278, 7745, 1092, 140,

2

(a) 140

(b) 2

(c) 7745

(d) 1092

(e) None of these

(19) 40, 73, 129, 203, 295, 405

(a) 40

(b) 405

(c) 295

(d) 203

(e) None of these

(20) 856, 761, 714, 692, 680, 675

(a) 714

(b) 761

(c) 692

(d) 675

(e) None of these

### Answers

(1) e

(2) d

(3) a

(4) b

(5) b

(6) e

(7) a

(8) b

(9) a

(10) c

(11) c

(12) d

(13) e

(14) b

(15) a

(16) b

(17) c

(18) c

(19) a

(20) c

### Solutions

(1) -62, -50, -40, -32, -26

-12, -10, -8, -6

(2)  $+6 \times 12.5$ ,  $+6 \times 14.5$ ,  $+6 \times 16.5$ ,

$+6 \times 18.5$ ,  $+6 \times 20.5$

(3)  $+14^2$ ,  $+17^2$ ,  $+20^2$ ,  $+23^2$ ,  $+26^2$

(4)  $+27 \times 2$ ,  $+28 \times 2$ ,  $+29 \times 2$ ,  $+30 \times 2$ ,

$+31 \times 2$

(5)  $+39$ ,  $+47$ ,  $+55$ ,  $+63$ ,  $+71$

$+8$ ,  $+8$ ,  $+8$ ,  $+8$

(6)  $-6^2+5$ ,  $-5^2+6$ ,  $-4^2+7$ ,  $-3^2+8$ ,  $-2^2+9$

(7)  $\times 2-16$ ,  $\times 3-18$ ,  $\times 4-20$ ,  $\times 5-22$ ,  $\times 6-24$

(8)  $\times 3+5$ ,  $\times 3+7$ ,  $\times 3+9$ ,  $\times 3+11$ ,  $\times 3+13$

(9)  $+28$ ,  $-32$ ,  $+36$ ,  $-40$ ,  $+44$

(10)  $\times 1^2+1$ ,  $\times 2^2+2$ ,  $\times 3^2+3$ ,  $\times 4^2+4$ ,  $\times 5^2+5$

(11)  $\times 2-8.5$ ,  $\times 2-10.5$ ,  $\times 2-12.5$ ,  $\times 2-14.5$ ,

$\times 2-16.5$

(12)  $\div 13$ ,  $\times 8$ ,  $\div 13$ ,  $\times 8$ ,  $\div 13$

(13)  $+800$ ,  $+400$ ,  $+200$ ,  $+100$ ,  $+50$

(14)  $-13^2$ ,  $-11^2$ ,  $-9^2$ ,  $-7^2$ ,  $-5^2$

(15)  $-11 \times 2$ ,  $-12 \times 3$ ,  $-13 \times 4$ ,  $-14 \times 5$ ,  $-15 \times 6$

(16)  $-4.5 \times 4$ ,  $-4.5 \times 4$ ,  $-4.5 \times 4$ ,  $-4.5 \times 4$ , -

$4.5 \times 4$

(17) Sum of the previous two

numbers

(18)  $\div 7-10$ ,  $\div 7-12$ ,  $\div 7-14$ ,  $\div 7-16$ ,  $\div 7-18$

(19)  $+38$ ,  $+56$ ,  $+74$ ,  $+92$ ,  $+110$

$+18$ ,  $+18$ ,  $+18$ ,  $+18$

(20)  $-95$ ,  $-47$ ,  $-23$ ,  $-11$ ,  $-5$

$-48$ ,  $-24$ ,  $-12$ ,  $-6$

## 5. MISSING NUMBER SERIES

1. ?, 703, 735, 713, 699, 691

- (a) 748
- (b) 777
- (c) 747
- (d) 744
- (e) 749

(2) 54, 18, 90, ?, 150, 50

- (a) 30
- (b) 35
- (c) 32
- (d) 31
- (e) 33

(3) ?, 110, 166, 208, 236, 250

- (a) 48
- (b) 47
- (c) 44
- (d) 45
- (e) 40

(4) 4, 15, 48, 147, 444, ?

- (a) 1230
- (b) 1320
- (c) 1220
- (d) 1330
- (e) 1335

(5) 10, 23, 75, ?, 1115, 4443

- (a) 288
- (b) 283
- (c) 285
- (d) 289
- (e) 298

(6) 8, 18, 32, 54, ?, 162

- (a) 92
- (b) 95
- (c) 99
- (d) 98
- (e) 88

(7) 30, ?, 130, 168, 198, 220

- (a) 84
- (b) 88
- (c) 85
- (d) 87
- (e) 78

(8) 163, 175, 191, 211, ?, 263

- (a) 231
- (b) 233
- (c) 230
- (d) 235

(e) 223

(9) 25, 125, ?, 645, 1129, 1805

(a) 325

(b) 324

(c) 322

(d) 320

(e) 321

(10) ?, 21, 43, 76, 120, 175

(a) 12

(b) 15

(c) 11

(d) 9

(e) 10

(11) 24, ?, 60, 96, 144, 204

(a) 34

(b) 35

(c) 36

(d) 33

(e) 30

(12) ?, 1652, 1527, 1302, 977, 552

(a) 1678

(b) 1657

(c) 1655

(d) 1677

(e) 1670

(13) 26, 197, ?, 559, 748, 941

(a) 377

(b) 375

(c) 370

(d) 379

(e) 378

(14) 29, 48.5, 68, 87.5, ?, 126.5

(a) 107

(b) 111

(c) 112

(d) 115

(e) 104

(15) 200, ?, 260, 320, 400, 500

(a) 220

(b) 225

(c) 228

(d) 234

(e) 235

(16) ?, 1726, 1530, 1274, 950, 550

(a) 1857

(b) 1875

(c) 1887

(d) 1870

(e) 1880

(17) 130, 125, ?, 104, 84, 55

(a) 101

(b) 104

(c) 112

(d) 117

(e) 100

(18) 33, 48, 35, 46, ?, 44

(a) 37

(b) 35

(c) 31

(d) 30

(e) 28

(19) 9, 12, 18, 30, 53, ?

(a) 89

(b) 98

(c) 94

(d) 87

(e) 88

(20) 88, ?, 187, 286, 473, 759

(a) 95

(b) 98

(c) 99

(d) 94

(e) 97

#### Answers

(1) c

(2) a

(3) e

(4) e

(5) b

(6) a

(7) a

(8) d

(9) e

(10) e

(11) c

(12) d

(13) b

(14) a

(15) a

(16) d

(17) d

(18) a

(19) c

(20) c

#### Solutions

(1) -44, -32, -22, -14, -8

-12, -10, -8, -6

(2)  $\div 3$ ,  $\times 5$ ,  $\div 3$ ,  $\times 5$ ,  $\div 3$

(3) +70, +56, +42, +28, +14

(4)  $\times 3+3$ ,  $\times 3+3$ ,  $\times 3+3$ ,  $\times 3+3$ ,  
 $\times 3+3$

(5)  $\times 4-17$ ,  $\times 4-17$ ,  $\times 4-17$ ,  $\times 4-17$ ,  
 $\times 4-17$

(6)  $+2^2+6$ ,  $+2^3+6$ ,  $+2^4+6$ ,  $+2^5+6$ ,  
 $+2^6+6$

(7) +54, +46, +38, +30, +22

-8, -8, -8, -8

(8)  $+4\times 3$ ,  $+4\times 4$ ,  $+4\times 5$ ,  $+4\times 6$ ,  
 $+4\times 7$

(9)  $+10^2$ ,  $+14^2$ ,  $+18^2$ ,  $+22^2$ ,  $+26^2$

(10)  $+2\times 5.5$ ,  $+4\times 5.5$ ,  $+6\times 5.5$ ,  
 $+8\times 5.5$ ,  $+10\times 5.5$

(11) +12, +24, +36, +48, +60

(12) -25, -125, -225, -325, -425

(13) +171, +178, +184, +189,  
+193

+7, +6, +5, +4

(14) +19.5, +19.5, +19.5, +19.5,  
+19.5

(15) +20, +40, +60, +80, +100

(16)  $-12^2$ ,  $-14^2$ ,  $-16^2$ ,  $-18^2$ ,  $-20^2$

(17) -5, -8, -13, -20, -29

+3, +5, +7, +9

(18) +15, -13, +11, -9, +7

(19) +3, +6, +12, +23, +41

+3, +6, +11, +18

(20) Sum of the previous two  
numbers

## 6. DATA INTERPRETATION

**SET 1** The bar graph shows the average number of students who appeared for SBI CLERK & RRB CLERK exam and difference between number of students who appeared for SBI CLERK & RRB CLERK exam. Read the data and answer the following questions.

बार ग्राफ़ SBI CLERK और RRB CLERK परीक्षा में बैठने वाले छात्रों की औसत संख्या और SBI CLERK और RRB CLERK परीक्षा में बैठने वाले छात्रों की संख्या के बीच अंतर दर्शाता है। डेटा पढ़ें और निम्नलिखित प्रश्नों के उत्तर दें।

| City | total number of students who appeared for (SBI Clerk+RRB Clerk) exam | percentage or number of students who appeared for RRB Clerk exam |
|------|----------------------------------------------------------------------|------------------------------------------------------------------|
| A    | 3600                                                                 | 44.44%                                                           |
| B    | 6300                                                                 | 42.84%                                                           |
| C    | 4500                                                                 | 46.66%                                                           |
| D    | 5700                                                                 | 2500                                                             |
| E    | 4400                                                                 | 1800                                                             |

1. The number of students who appeared for SBI CLERK exam in city B is what percent of number of students who appeared for RRB CLERK exam in city C?

शहर B में SBI CLERK परीक्षा देने वाले छात्रों की संख्या शहर C में RRB CLERK परीक्षा देने वाले छात्रों की संख्या का कितना प्रतिशत है?

- (A) 171.42%  
(B) 44.44%  
(C) 120%  
(D) 155.55%  
(E) None of these

2. What is the difference between number of students who appeared for RRB CLERK exam in city B and number of students who appeared for RRB CLERK exam in city E?

शहर B में RRB CLERK परीक्षा देने वाले छात्रों की संख्या और शहर E में RRB CLERK परीक्षा देने वाले छात्रों की संख्या के बीच क्या अंतर है?

- (A) 320  
(B) 470  
(C) 900  
(D) 670  
(E) None of these

3. The students who qualified the SBI CLERK exam and RRB CLERK exam in city E is 30.76% and 33.33% respectively. Then the total number of students who did not qualified the SBI CLERK exam and RRB CLERK exam in city E is what percent more or less than total number of students who appeared for SBI CLERK exam and RRB CLERK exam in city B?



शहर E में SBI CLERK परीक्षा और RRB CLERK परीक्षा उत्तीर्ण करने वाले छात्रों की संख्या क्रमशः 30.76% और 33.33% है। तो फिर शहर E में SBI CLERK परीक्षा और RRB CLERK परीक्षा उत्तीर्ण नहीं करने वाले छात्रों की कुल संख्या, शहर B में SBI CLERK परीक्षा और RRB CLERK परीक्षा देने वाले छात्रों की कुल संख्या से कितने प्रतिशत अधिक या कम है?

- (A) 52.38% less
- (B) 42.74% less
- (C) 41.42% less
- (D) 45.35% less
- (E) None of these

4. Find the average number of students who appeared for RRB CLERK exam in all five cities.

सभी पाँच शहरों में RRB CLERK परीक्षा देने वाले छात्रों की औसत संख्या ज्ञात कीजिए।

- (A) 2320
- (B) 2470
- (C) 2140
- (D) 2630
- (E) None of these

5. Find the ratio between number of students who appeared for RRB CLERK exam in city B and number of students who appeared for SBI CLERK exam in city C.

शहर B में RRB CLERK परीक्षा देने वाले छात्रों की संख्या और शहर C में SBI CLERK परीक्षा देने वाले छात्रों की संख्या के बीच अनुपात ज्ञात कीजिए।

- (A) 2 : 5
- (B) 9 : 8
- (C) 4 : 7

(D) 2 : 3

(E) None of these

### Solutions

| City | number of students who appeared for SBI Clerk exam | number of students who appeared for RRB Clerk exam | Total |
|------|----------------------------------------------------|----------------------------------------------------|-------|
| A    | 2000                                               | 1600                                               | 3600  |
| B    | 3600                                               | 2700                                               | 6300  |
| C    | 2400                                               | 2100                                               | 4500  |
| D    | 3200                                               | 2500                                               | 5700  |
| E    | 2600                                               | 1800                                               | 4400  |

1. (A) 171.42%

2. (C) 900

3. (A) 52.38% less { total number of students who did not qualified the SBI CLERK exam and RRB CLERK exam in city E =  $\frac{9}{13}$  of 2600 +  $\frac{2}{3}$  of 1800 = 3000 so answer =  $\frac{3300}{6300} \times 100 = 52.38\%$  less }

4. (C) 2140

5. (B) 9 : 8

**SET 2** The table shows the time taken by five person to complete the work on Monday(in min.) & the ratio between time taken on Monday to Wednesday and also shows the efficiency as a percentage of five persons on Tuesday with respect to Monday. Read the data and answer the following questions.

तालिका सोमवार को काम पूरा करने के लिए पांच व्यक्तियों द्वारा लिया गया समय (मिनट में) दिखाती है और सोमवार से बुधवार को लिए गए समय के बीच का अनुपात दिखाती है और सोमवार के संबंध में मंगलवार को पांच व्यक्तियों की दक्षता को प्रतिशत के रूप में दर्शाता है। डेटा पढ़ें और निम्नलिखित प्रश्नों के उत्तर दें।

| Person | Time taken to complete the work on Monday(in min.) | Ratio of Time taken(Monday : Wednesday) | efficiency as a percentage of these five persons on Tuesday with respect to Monday |
|--------|----------------------------------------------------|-----------------------------------------|------------------------------------------------------------------------------------|
| A      | 20                                                 | 04:05                                   | 25%                                                                                |
| B      | 35                                                 | 05:06                                   | 50%                                                                                |
| C      | 15                                                 | 03:04                                   | 30%                                                                                |
| D      | 25                                                 | 05:04                                   | 20%                                                                                |
| E      | 30                                                 | 03:05                                   | 40%                                                                                |

1. On Tuesday, A and B together worked for 30 minutes and then left. Find in how much time(approx.) C will complete the remaining work?

मंगलवार को, A और B ने एक साथ 30 मिनट तक काम किया और फिर छोड़ दिया। ज्ञात कीजिए कि C शेष कार्य कितने समय (लगभग) में पूरा करेगा?

(A)15 min

- (B)10 min
- (C)12 min
- (D)17 min
- (E)None of these

2. On Wednesday, another person F takes 25% more time than time taken by D and E to together complete the same work. Find the efficiency of the person F.

बुधवार को, एक अन्य व्यक्ति F को उसी कार्य को पूरा करने में D और E द्वारा लिए गए समय से 25% अधिक समय लगता है। व्यक्ति F की दक्षता ज्ञात कीजिए।

- (A)10
- (B)16
- (C)12
- (D)18
- (E)None of these

3. Find the ratio between time taken by B on Tuesday and Wednesday.

मंगलवार और बुधवार को B द्वारा लिए गए समय के बीच अनुपात ज्ञात कीजिए।

- (A)4 : 7
- (B)2 : 1
- (C)3 : 2
- (D)5 : 3
- (E)None of these

4. On Wednesday, B, C and D worked for 5 minutes and then B and C replaced by A and E and then the remaining work was completed by A, D and E working on alternatively started by D & followed by E and then A. Find the total time(approx.) in which total work was completed.

बुधवार को, B, C और D ने 5 मिनट तक काम किया और फिर B और C की जगह A और E ने ले ली और फिर शेष काम A, D और E द्वारा पूरा किया गया, वैकल्पिक रूप से D द्वारा शुरू किया गया और उसके बाद E और फिर A ने काम किया। . कुल समय (लगभग) ज्ञात करें जिसमें कुल कार्य पूरा हुआ।

- (A) 15 min  
(B) 10 min  
(C) 12 min  
(D) 17 min  
(E) None of these

5. The time taken by B on Monday is what percent of time taken by E on Wednesday?

B द्वारा सोमवार को लिया गया समय E द्वारा बुधवार को लिए गए समय का कितना प्रतिशत है?

- (A) 70%  
(B) 40%  
(C) 50%  
(D) 60%  
(E) None of these

Solutions

Hint : ratio given about time taken on Monday to the time taken on Wednesday so we can easily calculate the time taken on Wednesday.

Now it is given that bar graph shows the efficiency as a percentage of five persons on Tuesday with respect to Monday, so for person A ratio of efficiency on Monday to efficiency on Tuesday is  $4u : 1u$  so ratio of time taken on Monday : time taken on Tuesday will be  $1u : 4u$  and time taken on Monday is 20 min so  $1u = 20$  min and  $4u = 80$  min. So for person A time taken on Tuesday will be 80 min.

| Person | Time taken to complete the work on Monday(in min.) | Time taken to complete the work on Tuesday(in min.) | Time taken to complete the work on Wednesday(in min.) |
|--------|----------------------------------------------------|-----------------------------------------------------|-------------------------------------------------------|
| A      | 20                                                 | 80                                                  | 25                                                    |
| B      | 35                                                 | 70                                                  | 42                                                    |
| C      | 15                                                 | 50                                                  | 20                                                    |
| D      | 25                                                 | 125                                                 | 20                                                    |
| E      | 30                                                 | 75                                                  | 50                                                    |

1. (B) 10 min, A 80 35u

B 70 2800 40u

C 50 56u

So A & B together worked for 30 minutes, work done =  $(35+40) \times 30 = 2250$ .

Remaining work =  $2800 - 2250 = 550$ . So time taken by C to do this remaining work =  $550/56 = 9.82 = 10$  min approx.

2. (E) 5.6, D 20 5  
 100  
 E 50 2

So D+E total efficiency 7

Now it is given that person F takes 25% more time than time taken by D and E together

|                |     |    |
|----------------|-----|----|
| So ratio       | D+E | F  |
| Time           | 4u  | 5u |
| And efficiency | 5u  | 4u |

So  $5u = 7$  and  $1u = 1.4$ , so  $4u = 4 \times 1.4 = 5.6$ .

3. (D) 5 : 3

4. (A) 15 min

5. (A) 70%

**SET 3. The line graph shows the speed of bus A and bus B on five different days of the week and bar graph shows the percentage distribution of the total distance travelled by bus A and bus B on five different days of the week. Read the data and answer the following questions.**

लाइन ग्राफ सप्ताह के पांच अलग-अलग दिनों में बस ए और बस बी की गति दिखाता है और बार ग्राफ सप्ताह के पांच अलग-अलग दिनों में बस ए और बस बी द्वारा तय की गई कुल दूरी का प्रतिशत वितरण दिखाता है। डेटा पढ़ें और निम्नलिखित प्रश्नों के उत्तर दें।

| Day       | Speed of Bus A(kmph) | Speed of Bus B(kmph) | % of distance travelled by Bus A(km) | % of distance travelled by Bus B(km) |
|-----------|----------------------|----------------------|--------------------------------------|--------------------------------------|
| Monday    | 24                   | 28                   | 20%                                  | 25%                                  |
| Tuesday   | 32                   | 21                   | 22.50%                               | 17.50%                               |
| Wednesday | 25                   | 30                   | 28%                                  | 24.50%                               |
| Thursday  | 20                   | 22                   | 12.50%                               | 12%                                  |
| Friday    | 35                   | 40                   | 17%                                  | 21%                                  |

1. If the distance travelled by bus B on Thursday is 480 km then find the ratio between time taken by bus B on Monday to the time taken by bus B on Tuesday.

यदि गुरुवार को बस B द्वारा तय की गई दूरी 480 किमी है, तो सोमवार को बस B द्वारा लिए गए समय और मंगलवार को बस B द्वारा लिए गए समय के बीच का अनुपात ज्ञात कीजिए।

(A) 16 : 17

(B) 21 : 20

(C) 15 : 14

(D) 11 : 12



(E)None of these

2. If bus A travelled its journey on Thursday in 4.5 hours, then what is the difference between distance travelled by it on Monday and Tuesday?

यदि बस A ने गुरुवार को अपनी यात्रा 4.5 घंटे में तय की, तो सोमवार और मंगलवार को उसके द्वारा तय की गई दूरी के बीच क्या अंतर है?

(A)18 km

(B)20 km

(C)24 km

(D)32 km

(E)None of these

3. If bus A travelled 675 km on Tuesday, then what is the total time taken by the bus A to travel on Thursday & Friday together?

यदि बस A ने मंगलवार को 675 किमी की यात्रा की, तो गुरुवार और शुक्रवार को एक साथ यात्रा करने में बस A द्वारा लिया गया कुल समय कितना है?

(A)38(8/25) hours

(B)35(4/21) hours

(C)32(7/27) hours

(D)33(9/28) hours

(E)None of these

4. If bus B travelled 450 km on Monday, find the average distance travelled by bus A on Tuesday, Wednesday & Thursday. (Total distance of bus A is equals to total distance of bus B).

5. यदि बस B ने सोमवार को 450 किमी की यात्रा की, तो मंगलवार, बुधवार और गुरुवार को बस A द्वारा तय की गई औसत दूरी ज्ञात कीजिए। (Total distance of bus A is equals to total distance of bus B).

- (A)378 km  
(B)320 km  
(C)374 km  
(D)372 km  
(E)None of these

6. Between bus A & bus B, which bus travelled the less distance on Friday?

बस A और बस B के बीच, किस बस ने शुक्रवार को कम दूरी तय की?

- (A)bus A  
(B)bus B  
(C)both travelled the same distance  
(D)Can't be determined  
(E)None of these

Solutions

| Day       | Speed of bus A(in kmph) | Speed of bus B(in kmph) |
|-----------|-------------------------|-------------------------|
| Monday    | 24                      | 28                      |
| Tuesday   | 32                      | 21                      |
| Wednesday | 25                      | 30                      |
| Thursday  | 20                      | 22                      |

| Day       | % of distance travelled by bus A in km | % of distance travelled by bus B in km |
|-----------|----------------------------------------|----------------------------------------|
| Monday    | 20                                     | 25                                     |
| Tuesday   | 22.5                                   | 17.5                                   |
| Wednesday | 28                                     | 24.5                                   |
| Thursday  | 12.5                                   | 12                                     |

|        |    |    |
|--------|----|----|
| Friday | 35 | 40 |
|--------|----|----|

|        |    |    |
|--------|----|----|
| Friday | 17 | 21 |
|--------|----|----|

1. (C) 15 : 14, { distance travelled by bus B on Thursday is 480 km & from graph it is 12%, so  $12\% = 480$  and  $100\% = 4000$  km, so distance travelled by bus B on Monday = 25% of 4000 = 1000 km & distance travelled by bus B on Tuesday = 17.5% of 4000 = 700 km.

then time taken by bus B on Monday : time taken by bus B on Tuesday =  $1000/28$  :  $700/21 = 15 : 14$  .}

2. (A) 18 km
3. (D)  $33\frac{9}{28}$  hours
4. (A) 378 km
5. (D) Can't be determined { data is not sufficient to answer the que }

PRACTICE PAPER BY HASHISH ARORA

**SET 4. Study the data given below and answer the following questions.**

The data is about number of persons who likes three different types of water bottles. Number of persons who likes both Bisleri water and Aquafina water but not likes Clear waters is 150 which is 20% more than number of persons who likes both Clear waters and Bisleri water but not likes Aquafina water. The ratio of number of persons who likes only Aquafina water to number of persons who likes both Aquafina water and Bisleri water only is 7:5 respectively. Number of persons who likes all three types of water bottles is half of number of persons who likes only Clear waters. Number of persons who likes Clear waters is 581 while number of persons who likes both Clear waters and Aquafina water but not likes Bisleri water is 40 less than number of persons who likes all three types of water bottles. Number of persons who likes only Bisleri water is 42 more than number of persons who likes only Clear waters.

तीनों प्रकार की पानी की बोतलों को पसंद करने वाले व्यक्तियों के बारे में दिया गया डेटा इस प्रकार है। Bisleri water और Aquafina water दोनों पसंद करने वाले लेकिन Clear water पसंद नहीं करने वाले व्यक्तियों की संख्या 150 है, जो उन व्यक्तियों की संख्या से 20% अधिक है जो Clear water और Bisleri water दोनों पसंद करते हैं लेकिन Aquafina water पसंद नहीं करते। Only Aquafina water पसंद करने वाले व्यक्तियों और केवल Aquafina व Bisleri water पसंद करने वाले व्यक्तियों की संख्या का अनुपात क्रमशः 7 : 5 है। तीनों प्रकार की पानी की बोतलें पसंद करने वालों की संख्या, केवल Clear water पसंद करने वालों की संख्या का आधा है। Clear water पसंद करने वालों की कुल संख्या 581 है, जबकि Clear water और Aquafina water पसंद करने वाले लेकिन Bisleri water पसंद नहीं करने वाले व्यक्तियों की संख्या, तीनों प्रकार की पानी की बोतलों को पसंद करने वाले व्यक्तियों की संख्या से 40 कम है। केवल Bisleri water पसंद करने वालों की संख्या, केवल Clear water पसंद करने वालों की संख्या से 42 अधिक है।

1. Find the ratio between number of persons who likes all three types of water bottles and number of persons who likes Aquafina water & Clear water only.

तीनों प्रकार की पानी की बोतलों को पसंद करने वालों की संख्या तथा केवल Aquafina water और Clear water पसंद करने वालों की संख्या का अनुपात ज्ञात कीजिए।

(A) 31 : 21

(B) 21 : 29

(C) 22 : 23

(D) 39 : 41

(E) None of these

2. The number of persons who likes Aquafina water & Bisleri water only is what percent more or less than the number of persons who likes Bisleri water only?

केवल Aquafina water और Bisleri water पसंद करने वालों की संख्या, केवल Bisleri water पसंद करने वालों की संख्या से कितने प्रतिशत अधिक या कम है?

(A) 41.72% less

(B) 42.81% less

(C) 48.27% less

(D) 49.41% less

(E) None of these

3. Find the difference between number of person who likes Clear water only and number of person who likes Clear water and Aquafina water only.

केवल Clear water पसंद करने वाले व्यक्तियों और केवल Clear व Aquafina water पसंद करने वाले व्यक्तियों की संख्या में कितना अंतर है?

(A) 138

- (B)169
- (C)162
- (D)164
- (E)None of these

4. How many person likes all three types of water bottles?

कितने व्यक्ति तीनों प्रकार की पानी की बोतलों को पसंद करते हैं?

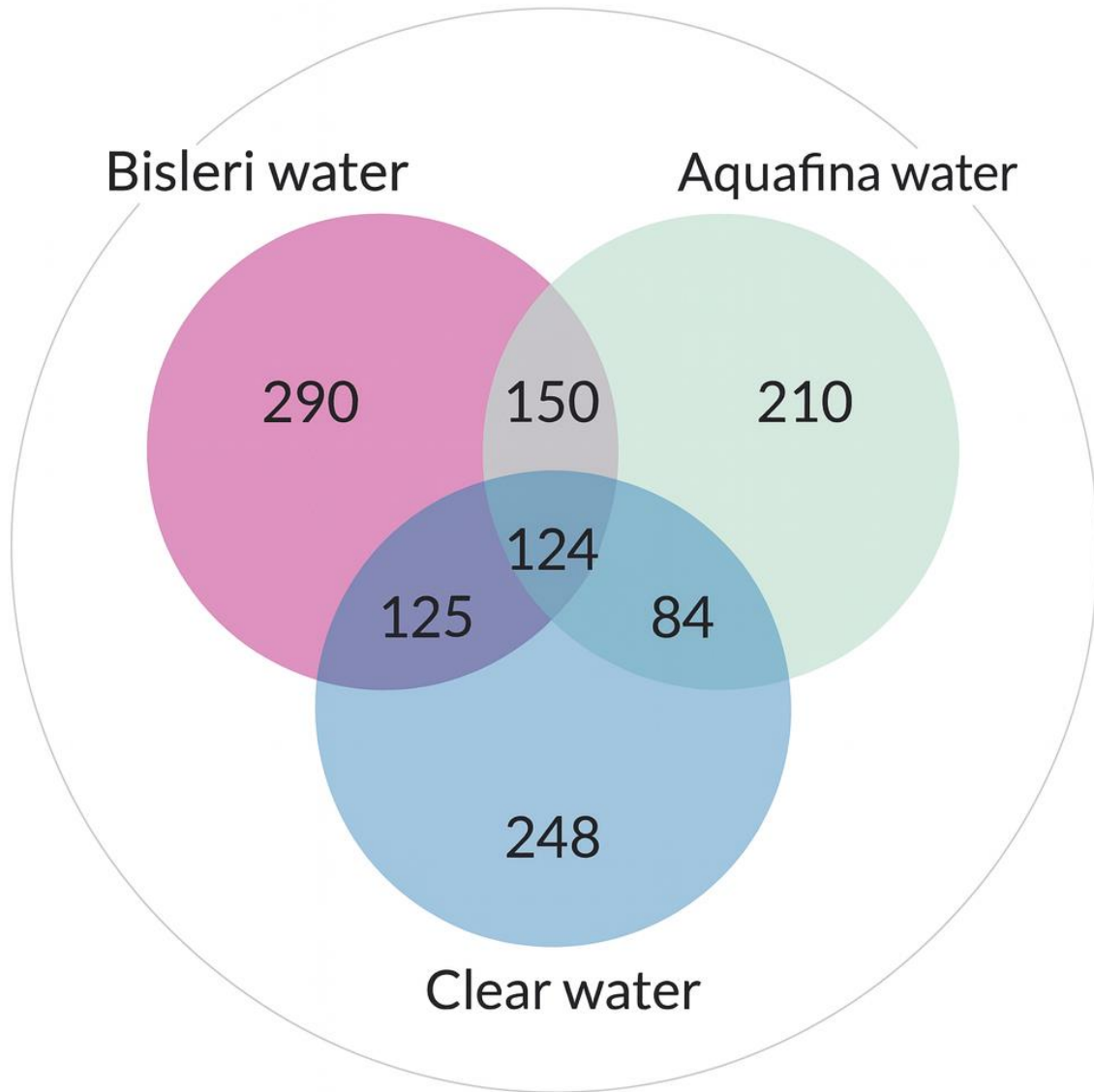
- (A)128
- (B)129
- (C)122
- (D)124
- (E)None of these

5. The number of person who likes Clear water and Bisleri water only is what percent of number of person who likes Aquafina water and Bisleri water only?

केवल Clear water व Bisleri water पसंद करने वालों की संख्या, केवल Aquafina water व Bisleri water पसंद करने वालों की संख्या का कितना प्रतिशत है?

- (A)81.22%
- (B)82.44%
- (C)83.33%
- (D)89.22%
- (E)None of these

## Solutions



1. (A) 31 : 21
2. (C) 48.27% less
3. (D) 164
4. (D) 124
5. (C) 83.33%

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